



Croatian Open Competition in Informatics

Round 3, December 13th 2025

Tasks

Task	Time limit	Memory limit	Score
Kist	1 second	512 MiB	50
Festival	1 second	512 MiB	70
Domjenak	1 second	512 MiB	110
Drzava	2 seconds	512 MiB	110
Ravnalo	1 second	512 MiB	110
Total			450



Task Kist

Ivo is in a square matrix of n rows and n columns where n is an odd number and holds a magic brush of thickness k . All cells of the matrix are marked with the symbol ".", and Ivo is in the central cell of the matrix.

Ivo has received a string of instructions, a word consisting of uppercase letters of the alphabet where each letter represents one instruction. He executes the instructions in order and does the following depending on the current instruction:



- L - Ivo moves one cell to the left
- R - Ivo moves one cell to the right
- U - Ivo moves one cell up
- D - Ivo moves one cell down
- If the current character is any other letter of the alphabet, Ivo **does not** change his position. Instead, he paints all the cells of the matrix that are * **less** than k cells away from his current position with the current color, regardless of whether they have already been painted in another color.

If Ivo moves outside the matrix, he will **skip** that step and continue executing the subsequent instructions. Your task is to print the appearance of the matrix after Ivo has executed all the instructions.

*The distance between two cells of a matrix is the minimum number of moves needed to get from one cell to the other, with movement allowed in four directions (up, down, left, right).

Input

The first line contains the natural numbers n, k ($1 \leq n, k \leq 50$), from the text of the task.

The second line contains a word consisting of uppercase letters of the English alphabet. The length of the word will be less than or equal to 50.

Output

In n lines, print n characters per line, the appearance of the matrix after Ivo executes all the instructions.

Scoring

Subtask	Points	Constraints
1	2	$n = 1$
2	10	$k = 1$
3	15	$k = 2$
4	23	No additional constraints.



Examples

input

1 1
ALURDF

output

F

input

3 2
LUUADDRCRB

output

AA.
ACB
CBB

Clarification of the first example:

Ivo will never move from the only cell of the matrix and will only paint that cell. After executing all the instructions, that cell is colored in color F.



Task Festival

While Ivan was working at this year's chocolate festival, his boss gave him n different chocolate candies and asked him to prepare k chocolate boxes using those candies. Before he starts arranging the boxes, Ivan wants to find out in how many different ways he can arrange them, and then, after considering all the possibilities, he will choose the best arrangement.

The chocolate festival is a very important and serious event, so the candies must be perfectly arranged. To achieve this, Ivan knows that he must follow the following rules:



- Each box must contain at least one candy.
- Each candy goes in exactly one box.
- The boxes he uses are identical to each other, so swapping the contents of two boxes does not create a new arrangement. All that matters is which candies are in which box and in what order they are arranged.
- The largest candy is always the first in the box. We can assume that in any group of candies, the largest can always be uniquely determined.

In how many different ways can Ivan arrange the boxes? Since the number of ways can be very large, print it modulo $10^9 + 7$.

Input

The first and only line contains the natural numbers n and k ($1 \leq n \leq 5000$, $1 \leq k \leq n$), the number of candies, and the number of boxes.

Output

In the first and only line, print a single number - the number of ways Ivan can arrange the boxes modulo $10^9 + 7$.

Scoring

Subtask	Points	Constraints
1	8	$k = 1$
2	19	$k = 2$
3	14	$n \leq 10$
4	29	No additional constraints.

Examples

input

3 1

output

2

input

3 2

output

3

input

4 2

output

11



Clarification of the first example:

Let's label the candies from smallest to largest with 1, 2, and 3. We need to arrange them in one box, so all three candies must be in the same box. Candy 3 must be first because it is the largest. We can arrange the other two candies in two ways, so $[3, 1, 2]$ and $[3, 2, 1]$ are the only valid arrangements.

Clarification of the second example:

Let's label the candies from smallest to largest with 1, 2, and 3. We can arrange the candies in three ways: $\{[1], [3, 2]\}$, $\{[2], [3, 1]\}$, and $\{[3], [2, 1]\}$.



Task Domjenak

Mr. Malnar is organizing a banquet for $2N$ of his associates that are numbered with integers $1, 2, \dots, 2N$. Associates came to the banquet in pairs. Looking at their behavior, Mr. Malnar noticed some strange rules that govern the spread of information within this group.

First, person A is willing to share information with person B if and only if they are friends. A pair of associates that came to the banquet together are friends.

Second, every person is willing to share information with at most one other person. At the same time, that person must not be aware of that information.

Third, every person will try to share information where priority is given to the person that came with them to the banquet.

In other words, if person A came with person B to the banquet and person A is aware of some information that person B is not, person A will share that information with person B. Because of the previous rules, person A will not share that information with anyone else and person B will try to share it with another person that they are friends with etc.

Fourth, the group can be split into two subgroups such that if two people are friends they are not in the same subgroup.

Mr. Malnar will tell his story to one of his associates and is interested in the maximum number of his associates that can hear his story. If that number is K , he is also interested in some sequence of associates $a_1, a_2, \dots, a_K$ such that if he is to tell his story to associate a_1 then for every $i = 1, 2, \dots, K - 1$ associate a_i can tell the story to associate a_{i+1} .

In the process of discovering said rules Mr. Malnar determined all friendships but forgot the pairs of associates that came to the banquet together. Fortunately, it is possible to recover this information. In other words, there is a unique partition of associates into N pairs such that associates within the same pair are friends.

Can you help Mr. Malnar answer his questions?

Input

The first line contains integers N and M ($1 \leq N \leq 5 \cdot 10^5, N \leq M \leq 10^6$), number of pairs of associates that came to the banquet and the number of friendships.

In each of the next M lines there are two integers u_i and v_i ($1 \leq u_i, v_i \leq 2 \cdot N$), labels of associates that are friends.

Output

In the first line, print a single integer K - the maximum number of associates that can hear Mr. Malnar's story.

In the second line, print a sequence of K integers that satisfy the constraints.





Scoring

Subtask	Points	Constraints
1	24	$N \leq 10$
2	16	Every associate is friends with at most two other associates.
3	30	$N \leq 2000, M \leq 5000$
4	40	No additional constraints.

Examples

input 2 3 1 2 1 4 3 4 output 4 2 1 4 3	input 3 5 1 2 2 3 1 4 4 5 1 6 output 4 6 1 2 3	input 4 8 1 2 2 3 3 4 4 1 2 5 3 6 4 7 7 8 output 6 6 3 4 1 2 5
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Clarification of the first example:

Associates that came together are: 1 and 2, 3 and 4.

Clarification of the second example:

Associates that came together are: 1 and 6, 2 and 3, 4 and 5. Longer sequences do not satisfy the constraints. For example, sequence (3,2,1,4,5) does not satisfy the constraints even though every two adjacent associates are friends because associate 1 can't tell the story to person 4 after hearing it from associate 2. This is because the associate that came with them to the banquet, namely associate 6, is still unaware of the story (third rule).

Clarification of the third example:

Associates that came together are: 1 and 4, 2 and 5, 3 and 6, 7 and 8.



Task Drzava

In a distant land called Airlanvaosma-i there are n cities connected by $n - 1$ roads. It is possible to reach every city from any other city by passing through fewer than 36 roads.

The almighty Krešimir wants to found his own state. Krešimir's state must have one capital city from which the state is governed, and several (possibly none) secondary cities that are under its administration. The size of a state is equal to the **total** number of cities that belong to that state (the capital city and all secondary cities).



However, in order for the governance of the state to be efficient, Krešimir imposes the following rule:

- for every secondary city, on the path from the capital city to that secondary city there must be **no** other secondary city. In other words, no secondary city may lie between the capital city and another secondary city.

For every state size k , where $1 \leq k \leq n$, determine how many different states of size k the almighty Krešimir can found. Since these numbers can be very large, output them modulo $10^9 + 7$.

Two choices of states are considered different if they differ in at least one chosen secondary city or in the choice of the capital city.

Input

The first line contains a natural number n ($1 \leq n \leq 3000$), the number of cities in the land.

In the next $n - 1$ lines there are two natural numbers u and v ($1 \leq u, v \leq n, u \neq v$), which indicate that cities u and v are connected. It will be possible to reach every city from any other city.

Output

In the first and only line output n numbers, the numbers described in the problem statement.

Scoring

Subtask	Points	Constraints
1	18	$1 \leq n \leq 15$
2	17	$1 \leq n \leq 200$
3	26	$1 \leq n \leq 600$
4	49	No additional constraints.



Sample tests

input

4
1 2
1 3
1 4

output

4 12 6 1

input

4
1 2
2 3
1 4

output

4 12 4 0

Explanation of the second sample test: There are 4 states of size 1 because there are 4 choices for the capital city. There are 4 states of size 2 because after choosing the capital city (4 ways), we also choose 1 secondary city (3 ways). There are 4 states of size 3 because by choosing capital cities 1 or 2 we have 2 ways each to choose 2 additional secondary cities.

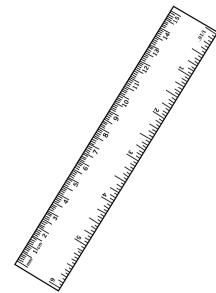


Task Ravnalo

Architect Hrvoje has been tasked with drawing a large irregular wall composed of vertical columns.

The wall consists of n columns placed next to each other, where the i -th column has height a_i and width 1. In order to make the drawing as complicated as possible, the i -th column is divided into b_i parts of equal height.

Hrvoje has only a ruler and a pencil at his disposal. With one stroke of the pencil he can draw one line segment (a straight line between two points) without lifting the pencil. His goal is to draw the entire wall, including all column edges and all boundaries between their divisions, using as few line segments as possible.



For the given wall, output the minimum number of line segments that need to be drawn in order to draw the wall completely.

Input

The first line contains a natural number n ($1 \leq n \leq 10^5$), the number from the problem statement.

The second line contains a sequence of n numbers $1 \leq a_i \leq 10^9$, from the problem statement.

The third line contains a sequence of n numbers $1 \leq b_i \leq 10^9$, from the problem statement.

Output

In the first and only line output one number – the minimum number of line segments Hrvoje needs to draw to draw the wall.

Scoring

Subtask	Points	Constraints
1	11	$N = 1$
2	13	$N = 2, 1 \leq a_i, b_i \leq 10$
3	29	$1 \leq a_i \leq 10^6, b_i$ divides a_i for every $1 \leq i \leq n$
4	57	No additional constraints.

Sample tests

input

3
4 6 4
2 3 4

output

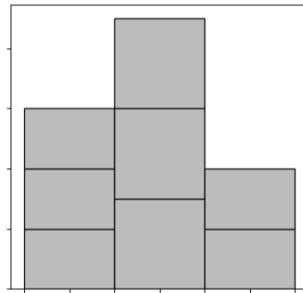
10

input

3
4 6 3
3 3 2

output

12



Pojašnjenje drugog probnog primjera: Hrvoje will extend the topmost line segment in the first column into the second column and thus “merge” 2 line segments into one. He will do the same with the 3 line segments at the bottom of the wall. This way, he will draw a total of 12 line segments, and it can be shown that this is the minimum number of line segments that Hrvoje needs to draw.